H-B-00481D(GSA-FSS)
October 15, 1970
Interim Revision of
Fed. Spec. H-B-481c(See 6.5)
August 31, 1965

INTERIM FEDERAL SPECIFICATION

BRUSHES, SANITARY

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of the Interim Federal Specification as a valid exception to Federal Specification H-B-481c, dated August 31, 1965.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers light duty and heavy duty sanitary brushes of various types and classes primarily used to clean receptacles such as toilet bowls, bedpans, and cuspidors (see 6.1).

1.2 Classification.

1.2.1 Duties. Brushes covered by this specification shall be of the following duties, based on the type of metal used in their construction, as specified (see 6.2):

Duty A - Light duty.
Duty B - Heavy duty.

1.2.2 Types and classes. Brushes covered by this specification shall be of the types and classes, as specified (see 6.2):

Type I - Twisted-in-wire curved brush head.

Class 1 - Tampico fiber.
Class 2 - Nylon filament.
Class 3 - Polypropylene filament.

Type II - Straight brush head, spiral wound.

Class 1 - Tampico fiber.

Type III - Straight brush head, twisted-in-wire.

Class 2 - Nylon filament.
Class 3 - Polypropylene filament.

Type IV - Ball brush head, wire-bound.

Class 4 - Palmetto fiber.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation to bid or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

H-S-951 - Synthetic Filaments, General Specifications for Brushes.
L-P-394 - Plastic Molding Material, Propylene Plastics, Injection and Extrusion.
L-P-1183- Plastic Molding Material, Acrylonitrile-Butadiene-Styrene (ABS) Rigid.
O-S-602 - Sodium Hypochlorite Solution.
O-S-809 - Sulfuric Acid, Technical.

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Federal Specifications:(con.)

QQ-N-290 - Nickel Plating, Electrodeposited.
QQ-S-698 - Steel, Sheet, and Strip, Low Carbon.
QQ-S-766 - Steel Plates, Sheets, and Strips, Corrosion Resisting.
QQ-W-423 - Wire, Steel, Corrosion Resisting.
QQ-W-461 - Wire, Steel, Carbon (Round, Bare, and Coated).
PPP-B-636 - Box, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM) Standards:

B 36 - Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar.
D 785 - Rockwell Hardness of Plastics and Electrical Insulating Materials.
D 2016 - Test for Moisture Content of Wood.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Commercial Standards:

CS 35-61 - Hardwood Plywood.

(Copies of Commercial Standards, which are Department of Commerce publications, may be obtained upon application accompained by money order, coupon, or cash to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

National Motor Freight Traffic Association Inc., Agent:

National Freight Classification Committee:

(Application for copies should be addressed to the National Motor Freight Traffic Association Inc., Agent, ATTN: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

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3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, three sample brushes of each duty, type, and class shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of the responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Materials.

3.2.1 Fiber and filament.

3.2.1.1 Class 1, tampico fiber. The fiber in class 1 brushes shall be natural color (undyed), unbleached, tula tampico fiber.

3.2.1.2 Class 2, nylon filaments. Class 2 filaments shall comply with the requirements for class N of H-S-951. The filaments shall be either round in cross section and crimped in accordance with type III of H-S-951, or X-shaped in cross section (see figure 5), at the option of the manufacturer. The average diameter of crimped filaments shall be 0.012 ± 0.001 inch (see 4.4.1.3), the average crimp frequency No. 13, and the average wave amplitude HD (deep) 0.010 inch (see 4.4.1.5). The average diameter of X-shaped filaments shall be 0.015 ± 0.001 inch (see 4.4.1.4). The filaments shall be natural color (undyed), or black at the option of the manufacturer.

3.2.1.3 Class 3, polypropylene filaments. Class 3 filaments shall comply with the requirements for class P of H-S-951. The filaments shall be either round in cross section and crimped in accordance with type III of H-S-951, or X-shaped in cross section (see figure 5), at the option of the manufacturer The average diameter of crimped filaments shall be 0.012 ± 0.001 inch (see 4.4.1.3), the average crimp frequency No. 10, and the average wave amplitude 0.022 inch (see 4.4.1.5). The average diameter of X-shaped filaments shall be 0.015 ± 0.001 inch (see 4.4.1.4). The filaments shall be natural color (undyed) or black, at the option of the manufacturer.

3.2.1.4 Class 4, palmetto fiber. The fiber in type IV brushes shall be oiled, natural (undyed) palmetto fiber.

3.2.2 Wood. Wood in sanitary brushes (see 3.4.1.2) shall be close-grain and straight-grain hardwood, medium or high density, such as beech, birch, maple, oak, cherry, elm, hickory, or pecan as classified in Commercial Standard CS-35, or ramin (Gonystylus spp.) (see 6.4). The wood shall be free from defects which may impair its serviceability or mar its appearance (see 4.3.3.2).

3.2.2.1 Seasoning. Wood blocks and handles shall be fabricated from wood which has been uniformly dried and maintained at a moisture content of 6 to 12 percent (see 4.3.2).

3.2.3 Plastics. Plastics for handles of type I brushes shall be one of the following materials at the option of the manufacturer:

(1) Polypropylene in conformance with type I of L-P-394 except that the physical requirements shall be limited to density, yield strength, and impact strength therein. In addition, polypropylene shall have a Rockwell hardness (R scale) of 96 to 104, inclusive, when tested as specified in D 785 (see 4.3.1 and 4.4.8).

(2) Acrylonitrile-butadiene-styrene in conformance with L-P-1183 except that the physical requirements shall be limited to specific gravity, impact strength, tensile yield stress, and Rockwell hardness (R scale) (see 4.3.1 and 4.4.8).

3.2.3.1 Auxiliary adaptors. Auxiliary adaptors shall be molded plastic, identical or compatible with the material in the handle.

3.2.4 Metal parts. Twisting wire and winding wire, ferrules, nails, and hooks used in the fabrication of sanitary brushes shall be as specified in 3.2.4.1 through 3.2.4.4. The temper of metal parts shall be the manufacturer's option provided the metal parts otherwise meet the requirements herein.

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3.2.4.1 Wire in duty A brushes. Wire in light-duty brushes shall be basic steel AISI 1006 or 1008, with finish 5, class 1 or 2 in accordance with QQ-W-461. The diameter and gage of the wire for each brush type shall be as specified herein in tables I, II, III, or IV, as applicable.

3.2.4.2 Wire in duty B brushes. Wire in heavy-duty brushes shall be corrosion resisting steel AISI 304 or 316 in accordance with QQ-W-423. The diameter of the wire for each brush type shall be as specified herein in tables I, II, III, or IV, as applicable.

3.2.4.3 Metal ferrules, hooks, nails, and staples in duty A brushes. Ferrules of light-duty brushes shall be corrosion resisting steel, nickel-plated steel, or nickel-plated brass at the option of the manufacturer. Plated steel shall be low carbon AISI 1006 or 1008 in accordance with QQ-S-698. Plated brass shall be alloy 260 in ASTM B 36. The coating shall be electrodeposited nickel in accordance with QQ-N-290, class 2. The thickness of the ferrule shall be not less than 0.010 inch. Nails, staples, and hooks shall be zinc coated.

3.2.4.4 Metal ferrules, hooks, nails, and staples in duty B brushes. Ferrules of type I heavy-duty brushes shall be corrosion resisting steel AISI 304 or 306 in accordance with QQ-S-766. The thickness of the ferrule shall be not less than 0.010 inch. Nails, staples, and hooks shall also be corrosion resisting steel.

3.2.5 Bonding agent for type I brushes.

3.2.5.1 Brushes with wood handles. The bonding agent for securing the twisted wire ends of the brush head within the wood handle shall be an adhesive such as an epoxy resin that is water and chemical resistant (see 3.8.1 and 3.8.2).

3.2.5.2 Brushes with plastic handles. The bonding agent for securing the twisted wire ends of the brush head within the plastic handle, or an auxiliary adaptor and collar to the handle, shall be (1) the plastic material of the handle or adaptor fused to the wire ends by heat induction, (2) a solvent-type adhesive, or (3) an epoxy resin, provided the resulting bond between joined parts is resistant to water and chemicals (see 3.8.1 and 3.8.2).

3.2.6 Varnish or lacquer. The coating applied to wood handles shall be a water and chemical resistant clear varnish or lacquer (see 3.8.2).

3.3 Design. Sanitary brushes shall be of the general designs shown in figures 1 through 4. The illustrations are included for convenience in identifying each brush type and are not intended to preclude designs otherwise meeting the requirements of this specification.

3.4 Construction. Sanitary brushes shall be constructed as follows and shall conform with the physical requirements in tables I through IV, as applicable.

3.4.1 Type I, twisted-in-wire curved brush head.

3.4.1.1 Brush head. Type I brush heads shall be horse-collar-shaped and contoured to reach the inside surfaces of wash basins, toilet bowls, bedpans, and the like, as illustrated in figure 1. The fibers or filaments shall be evenly and closely distributed in the twists of two wires. The fibers or filaments shall project perpendicular to the wire in the brush head and flare around the circumference of the wire forming a sufficiently dense working surface over the entire brush head to meet the test in 4.4.2. The fibers or filaments shall be evenly trimmed.

3.4.1.2 Handle. Unless otherwise specified, handles of type I brushes shall be made of wood specified in 3.2.2 or plastics specified in 3.2.3, at the option of the manufacturer.

3.4.1.3 Wood handle. Wood handles shall be straight, circular in cross section, and of the same diameter throughout. One end of the handle shall have a metal ferrule as specified in 3.2.4.3 or 3.2.4.4, as applicable, staked to the handle and extending over the edge and covering not less than 1/16 inch around the circumference of the cross section. The same end shall have a center hole of sufficient size to receive not less than 2-1/2 inches of the twisted wire ends of the brush part and the adhesive specified in 3.2.5.1. The adhesive shall be applied to the wire ends and allowed to thoroughly dry prior to assembly to assure that the entire circumference of each wire to be inserted in the handle is covered by the adhesive. The holes shall be filled with the adhesive, and the

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adhesive-covered wire ends passed through the collar and ferrule down into the hole to a depth of not less than 2-1/2 inches. The opposite end of the handle shall be rounded and shall be provided with a means of hanging the brush in accordance with 3.4.5. The handle shall be coated with a clear waterproof finish as specified in 3.2.6, before or after assembly.

3.4.1.4 Plastic handles. Plastic handles of type I brushes shall be molded plastic in accordance with 3.2.3. The shape of the handle shall be in accordance with the manufacturer's commercial practice provided it does not conflict with the basic requirements for plastic handles herein. One end of the handle shall have (1) a center hole, (2) an integral adaptor, or (3) an auxiliary adaptor and collar of the same material or a plastic material compatible with the material of the handles. Auxillary adaptors shall be bonded to the handle by a suitable bonding agent as specified in 3.2.5.2. The hole in the handle or in the adaptor, as applicable, shall be of sufficient size to receive not less than 1-1/4 inches of the twisted wire ends of the brush part plus the bonding agent.

TABLE I. Physical requirements of type I brushes

Characteristics		Requirements	
		With wood handle 1/	With plastic handle 1/
Handles:			
Length	inch	14 ± 1/4	14 ± 1/4
Cylindrical, diameter	do.	7/8 ± 1/8	7/8 ± 1/8
Rectangular or oval:			
Width at narrowest dimension, minimum	inch	-	1/2
Width at widest dimension, minimum	do.	-	1
Depth (overall thickness), minimum	do.	-	1/2
Ferrule, metal:			
Depth, minimum	inch	7/16	-
Overlap (collar), minimum	do.	1/16	-
Thickness, minimum	do.	0.010	-
Auxiliary adaptors:			
Depth of collar, minimum	inch	-	3/4
Brush part:			
Length, overall	inches	5-1/2 ± 1/4	5-1/2 ± 1/4
Width, overall			
At widest dimension	inches	4-1/2 ± 1/4	4-1/2 ± 1/4
At ferrule	do.	2-1/4 ± 1/4	2-1/4 ± 1/4
Periphery, overall	do.	13-1/2 ± 1/2	13-1/2 ± 1/2
Trimmed diameter of filaments, minimum	do.	1-7/8	1-7/8
Fiber and filament:			
Weight per brush, minimum			
Class 1 - Tampico	ounce	1.75	1.75
Class 2 - Nylon	do.	1.50	1.50
Class 3 - Polypropylene	do.	1.25	1.25
Weight per 1-inch length of X-shaped filament, average			
Class 2 - Nylon	gram	0.00195 ± 0.0001	0.00195 ± 0.0001
Class 3 - Polypropylene	do.	0.00190 ± 0.0001	0.00190 ± 0.0001
Wire, diameter (U. S. Steel Gage No. 13)	inch	0.092 ± 0.003	0.092 ± 0.003

1/ Unless otherwise specified (see 6.2), handles shall be wood or plastic at the option of the manufacturer.

3.4.2 Type II, straight brush head, spiral wound.

3.4.2.1 Brush part. The brush part, consisting of tampico fibers specified in 3.2.1.1 shall be formed at one end of a straight piece of wood specified in 3.2.2. The wood shall be circular in cross section and of the same diameter over the brush portion except for spiral grooves around the circumference to retain the fibers and an undercut portion at the dome end. The fibers shall be evenly

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and closely distributed and secured at their centers in the spiral grooves by the winding wire specified in 3.2.4.1 or 3.2.4.2, as applicable. The end of the wire near the handle shall be anchored to the handle by a nail (see 3.2.4.3 or 3.2.4.4). The fibers shall project perpendicular to the wood and flare around the circumference of the wood forming a sufficiently dense working surface over the entire brush head to meet the test in 4.4.2. The domed end shall consist of a small wire-bound tuft as shown in figure 2, formed on the first spiral or on an undercut portion of the wood, located so that the fibers extend beyond the end of the wood not less than 3/4 inch, and bound to the wood close to the base of the tuft with the winding wire. The fibers or filaments shall be evenly trimmed.

3.4.2.2 Handle. The handle end of the wood shall be circular in cross section and of the same diameter throughout as the wood in the brush part, or be slightly curved beginning at a distance of 1-1/4 inch from the brush part. The end of the handle shall be provided with means for hanging the brush as specified in 3.4.5 and shall be finished with a clear coating as specified in 3.2.6.

TABLE II. Physical requirements for type II brushes

Characteristic		Requirements
Handle:		
Length from base of brush part	inches	16 + 1, -0
Diameter of wood in handle and brush part	inch	15/16 ± 1/16
Brush part:		
Width	inches	4 ± 1/4
Length	do.	5-1/2 ± 1/4
Fiber:		
Length from handle, minimum	inches	1-1/2
Weight per brush, minimum	ounce	1.96
Wire, diameter (U. S. Steel Gage No. 21)	inch	0.031 ± 0.001

3.4.3 Type III, straight brush head, twisted-in-wire.

3.4.3.1 Brush part. Type III brushes shall have a straight brush head composed of the filaments specified in 3.2.1.2 or 3.2.1.3, as specified, securely held at their centers between the twists of two wires one strand doubled). The filaments shall project perpendicular to and flare around the circumference of the wires in the brush part, ending in a fan-shaped tip as shown in figure 3, formed by turning back the twisted ends and recessing them in the brush part. The fibers or filaments forming the working surface shall be sufficiently dense to meet the test in 4.4.2, and shall be evenly trimmed.

3.4.3.2 Handle. The handle of type III brushes shall be a continuation of the wires in the brush part, offset from the brush part as shown in figure 3. The wires in the handle shall be twisted so that there are not less than five and not more than 8 twists per inch. The closed end of the doubled wire strand shall be formed into an integral loop 2-3/4 ± 1/8 inches long and 1 ± 1/8 inch wide for convenience in holding the brush and also for hanging it when it is not in use.

TABLE III. Physical requirements for type III brushes

Characteristic		Requirement
Length, overall	inches	17-1/2 ± 1/2
Brush part:		
Length, overall (including fan-shaped end)	inches	6-1/2 ± 1/2
Diameter (trimmed)	do.	2 + 1/4, -0
Handle:		
Length to offset from handle end	inches	9 + 1/2, -0
Length to offset from base of brush part, minimum	inch	3/4
Offset	do.	1 + 1/4, -0

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TABLE III. Physical requirements for type III brushes (con.)

Characteristic		Requirement
Wire, diameter (U. S. Steel Gage No. 12)	inch	0.106 ± 0.003
Filament:		
Weight per brush, minimum		
Nylon, class 2	ounce	1.50
Polypropylene, class 3	do.	1.25
Weight per one-inch length of X-shaped filament, average		
Nylon, class 2	gram	0.00195 ± 0.0001
Polypropylene, class 3	do.	0.00190 ± 0.0001
Wire, diameter, (U. S. Steel Gage No. 12)	inch	0.106 ± 0.003

3.4.4 Type IV brushes, ball-head, wire-bound.

3.4.4.1 Brush part. The brush part of ball head brushes shall be formed at one end of a straight piece of wood circular in cross section as specified in 3.2.2. The brush portion of the wood shall have grooves around the circumference, spaced and shaped to aid in forming a ball-shaped brush head as shown in figure 4. The brush part shall consist of overlapping tufts of class 4 palmetto fiber as specified in 3.2.1.4, formed by securing the fibers at their centers in the grooves with the winding wire specified in 3.2.4.1 or 3.2.4.2, as applicable. In addition, each tuft shall be bound to the wood by wrapping close to the base of the tuft with winding wire. The wire shall be fastened to the handle in two places, one at the end of the wire using a No. 16 nail (see 3.2.4.3 or 3.2.4.4, as applicable), and the other diametrically opposite the first nail, using a nail staple, to prevent the wire from slipping off the fibers, (see 3.2.4.3 or 3.2.4.4, as applicable). The nail lengths shall not exceed the diameter of the handle.

3.4.4.2 Handle. The handle portion of the straight piece of wood shall be circular in cross section and of the same diameter throughout as the wood in the brush part or it may be tapered to a beaded end. The handle shall be finished as specified in 3.2.6 prior to assembly of the brush. The end of the handle shall be provided with a means for hanging the brush as specified in 3.4.5.

TABLE IV. Physical characteristics for type IV brushes

Characteristic		Requirement
Length of wood, overall	inches	14-1/2 ± 1/2
Diameter of wood, even		11/16 ± 1/32
Handle (tapered)		
Length out of brush part	do.	12 ± 1/4
Diameters:		
At the brush part	inch	11/16 ± 1/32
Approximately 3/8 inch from brush part	do.	7/16 ± 1/32
Bead end of block	do.	9/16 ± 1/32
Brush part:		
Diameter, approximate	inches	3-3/4
Radius of end trim, approximate	inch	1-3/4
Length	inches	4-1/4
Fiber:		
Weight per brush, minimum	ounces	2.88
Wire, diameter (U. S. Steel Gage No. 21)	inch	0.031 ± 0.001

3.4.5 Provision for hanging brushes. Types I, II, and IV brushes shall be provided with a convenient means for hanging them when not in use (see 6.3), such as, (1) a hole centrally located near the end of the handle, (2) a hook made of metal as specified in 3.2.4.3 or 3.2.4.4, as applicable or (3) an adaptor with hole or attached metal hook, as specified in 3.2.4.3 or 3.2.4.4, as applicable. The size and location of the hole shall be the manufacturer's option provided the serviceability, durability, or appearance of the brush is not adversely affected.

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3.5 Color of handles.

3.5.1 Wood handles. Wood handles shall be natural color.

3.5.2 Plastic handles, type I. The color of plastic handles shall be the manufacturer's option.

3.6 Properties of fibers and filaments.

3.6.1 Weight.

3.6.1.1 Total weight of fibers or filaments in each brush. The total weight of fibers or filaments in each brush shall be as specified in tables I through IV when determined as specified in 4.4.1.1 or 4.4.1.2.1, as applicable.

3.6.1.2 Average weight per inch length of X-shaped filaments. The average weight per inch length of X-shaped filaments shall be as specified in tables I and III when determined as specified in 4.4.1.2.2.

3.6.2 Specific gravity and melting point of classes 2 and 3 filaments. The specific gravity and melting point of class 2 nylon and class 3 polypropylene filaments shall be as specified in H-S-951 when determined as specified in 4.4.1.6 herein.

3.6.3 Resistance of fibers and filaments to boiling water. When tested for one hour in boiling water, 100° ± 1.1° C. (212° ± 2° F.) at sea level, as specified in 4.4.1.7, fibers or filaments shall show no evidence of artificial color, or in the case of black synthetic filaments, no bleeding, and after drying, no permanent change in stiffness.

3.7 Tuft retention. When subjected to a pulling force of 8 ± 0.5 pounds as specified in 4.4.3, no tuft shall be dislodged from the brush head.

3.8 Resistance of brushes to chemicals.

3.8.1 After 24-hour immersion. When immersed in solutions as specified in table V, by the method described in 4.4.4.1, brushes shall meet the following requirements:

(1) The bonding agent shall show no signs of softening or deteriorating in any way that will permit loosening or separation of bonded parts.
(2) Fibers or filaments shall show no evidence of artificial color, or in the case of black synthetic filaments, no bleeding.

TABLE V. Immersion solutions

Brush	Solution	Concentration of solution	Period of immersion	Temperature of solution	
		percent by wt.	hours	°C	°F.
Duty A	Sodium hypochlorite[1]	1.0 available chlorine	24	23° ± 2°	73.5° ± 3.6°
Duty B	Sodium hypochlorite[1]	1.0 available chlorine	24	23° ± 2°	73.5° ± 3.6°
	Sulfuric acid[2]	5.0	24	23° ± 2°	73.5° ± 3.6°

1. Prepared from sodium hypochlorite conforming to O-S-602.
2. Prepared from sulfuric acid conforming to class 2, type II of O-S-809.

3.8.2 After drying. After drying as specified in 4.4.4.1, brushes shall meet the following requirements.

(a) Fibers or filaments shall show no evidence of damage or change in stiffness that will affect the serviceability of the brush.
(b) Wire and other metal parts shall be free from rust and otherwise unaffected.
(c) The bonding agent shall not become brittle or deteriorate in any way that permits loosening or separation of bonded parts or loosening of wire twists and shedding of fibers or filaments before or after testing as specified in 4.4.5, 4.4.6 and 4.4.7.
(d) Handles shall not crack, split, break, chip, or craze, as applicable, and coatings on wood handles shall not become soft, peel, or chip before or after testing as specified in 4.4.7.
(e) Wire twists shall not loosen and permit shedding of fibers or filaments, or loss of tufts, before or after testing as specified in 4.4.5, 4.4.6 or 4.4.7.

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3.9 Identification marking. Types I, II, and IV brushes shall be marked with the manufacturer's name or trademark and Federal stock number. In addition, type I brushes with plastic handles shall be marked or tagged with the name of the plastic material in the handle. Type III brushes shall be tagged with the manufacturer's name or trademark and Federal stock number.

3.10 Workmanship. Brushes shall be free from defects which may affect their serviceability, durability, or appearance. Occurrence of defects shall not exceed the applicable quality level (AQL).

4. QUALITY ASSURANCE PROVISIONS

4.1 Reponsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the examination and testing set forth in the specification where such are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required, it shall be examined for defects in table VI and for specified dimensions. The preproduction sample shall be tested in accordance with 4.4, as applicable to each duty, type, or class.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Inspection of components and materials. In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with the specified requirements or referenced subsidiary specifications and standards or if none, in accordance with this specification.

4.3.2 Intermediate testing. Tests shall be performed on each lot of wood offered for acceptance at time of fabrication for moisture content in accordance with method B of ASTM D2016. The sample unit shall be one piece of wood. Three determinations shall be taken on each sample unit and results shall be the average of the three determinations. The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.5 defects per hundred units.

4.3.3 End item examination and testing.

4.3.3.1 Lot. For the purpose of examination and testing of the end item, a lot shall consist of all brushes of one duty, type, and class offered for delivery at one time.

4.3.3.2 Visual examination. Brushes shall be examined visually for defects in table VI. The inspection level shall be II and the acceptable quality level (AQL) 4.0 defects per 100 units. The unit for examination shall be one brush. If the number of defects in the sample for visual examination exceeds the specified acceptance number in MIL-STD-105, the entire lot of brushes shall be considered to be defective.

4.3.3.3 Dimensional examination. Brushes shall be examined for conformance with the applicable dimensional requirements in tables I through IV. The inspection level shall be S-3, and the acceptable quality level (AQL) 6.5 percent defective. The unit for examination shall be one brush. If any brush in the sample for test fails to meet any of the applicable requirements in tables I through IV for which examination is made, the brush shall be considered defective. If the number of defective brushes in the sample for examination exceeds the acceptance number in MIL-STD-105, the entire lot of brushes shall be considered defective.

TABLE VI. Classification of defects

Examination	Defect
Material Wood in types I, II, and III	Not dense close-grain hardwood free from cracks, splits, checks, knotholes, or firmly encased knots 3/16 inch or more in diameter. Warped (visible deformation).
Plastic handles, type I, style B	Not as specified.
Synthetic filaments	Not as specified.
Vegetable fibers	Not as specified.
Wire, nails, hooks, and ferrules	Not as specified.

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TABLE VI. Classification of defects (con.)

Examination	Defect
Construction and design.	
General defects applicable to all brushes:	
Design	Shape not as specified.
Handles	No provision for hanging.
Brush part	Fibers or filaments not evenly trimmed and evenly distributed in brush part; not fully flared as specified.
	Not clean; evidence of foreign matter.
Wire	Scratched or nicked.
Zinc coating on wire	Basic wire completely exposed in some areas; coating flakes when rubbed with fingers.
Type I. Twisted-in-wire, curved brush head	
Bonding agent	Adhesive brittle - can be removed and reduced to powder by crushing with knife blade.
	Does not fill hole in handle or adaptor, as applicable.
	Joined parts loose and easily moved or separated by hand.
Finish on handle	Not clear as specified.
	Unevenly applied.
	Coating absent.
Metal ferrule or auxiliary adaptor, as applicable	Not securely fastened to handle.
	Metal ferrule does not extend over edge of handle as specified.
Fibers or filaments	Not securely held in wire twists. Can be easily removed by hand.
Type II. Spiral wound straight brush head	
Tampico fiber	Not securely bound to wood with winding wire.
	Fiber unevenly distributed in spiral grooves.
Wire	Wire ends not anchored to handle.
Nails	Nail ends exposed.
Dome end of brush head	Absent.
	Not formed as specified.
Type III. Twisted-in-wire straight brush head	
Handle	Not offset from brush part.
	Less than 5 or more than 8 twists per inch in handle.
	Twists not formed by two strands of wire (one strand doubled).
	Loop at end of handle not as specified.
Brush part	Filaments or fibers not securely held in wire twists. Can be easily removed by hand. Wire ends not recessed in brush part.
Type IV. Wire-bound ball head brush	
Brush part	Fibers not securely bound to block with winding wire.
	Wire not securely fastened to handle as specified.
	Nail ends protrude.
Marking, identification (see 3.9)	Illegible, incomplete.

4.3.3.4 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for delivery at one time. The inspection level shall be S-2 and the AQL shall be 6.5 defects per hundred units. Defects shall be scored as specified in table VII.

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TABLE VII. Examination of preparation for delivery

Examination	Defect
Markings (exterior and interior containers)	Missing, incorrect, illegible, or improper size, location, sequence, or method of application.
Material	Component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of case liners (when applicable), container flaps, loose strapping or tape banding; bulging or distortion of containers.
	Intermediate package containers more or less than required.

4.3.3.5 Testing. Brushes shall be tested for conformance with the applicable requirements of this specification by the methods specified in 4.4. The test unit shall be three duty A brushes (or four duty B brushes, as applicable). The inspection level shall be S-2 and the acceptable quality level (AQL) 4.0. If any brush in the sample for test fails to meet any requirements for which test is made, the test unit shall be considered defective. If the number of defective test units in the sample for test exceeds the specified acceptance number in MIL-STD-105, the entire lot of brushes shall be considered defective.

4.4 Test methods. Unless otherwise specified herein, tests shall be conducted at room temperature, $27^{\circ} \pm 5^{\circ}$ C. ($80^{\circ} \pm 9^{\circ}$ F.).

4.4.1 Fibers and filaments.

4.4.1.1 Weight of vegetable fibers. Fibers from one brush of class 1 tampico or class 4 palmetto brush, as applicable, shall be tested as follows: Remove the fibers from the brush and condition for not less than 16 hours in air at a temperature of $23^{\circ} \pm 1.1^{\circ}$C. ($73.5^{\circ} \pm 2^{\circ}$F.) and a relative humidity of 50 ± 4 percent. At the end of the conditioning period, weigh the fibers in the same atmosphere and check the results for conformance with the requirements in tables I, II, or IV, as applicable.

4.4.1.2 Weight of synthetic filaments.

4.4.1.2.1 Total weight of filaments per brush. Filaments from one brush of class 2 nylon or from one brush of class 3 polypropylene shall be tested as follows: Remove the filaments from the brush, weigh, record, and check the result for conformance with the requirement in tables I or III, as applicable. Reserve the filaments for use in the tests in 4.4.1.2.2 through 4.4.1.7.

4.4.1.2.2 Average weight per inch length of X-shaped filaments. Take 20 filaments from the weighed filaments in 4.4.1.2.1 reserved for this purpose. Cut a one-half inch length from each filament and weigh the 20 one-half inch lengths on an analytical balance accurate to $\pm$ 0.0001 gram. Calculate the weight per one-inch length as follows, and check the result for conformance with the applicable requirement in table I or III.

$$A = \frac{B}{C}$$

where: A = Average weight of one-inch length of filament, grams.
B = Total weight of 20 one-half inch lengths of filament, grams.
C = Number of one-inch lengths, or 10.

4.4.1.3 Diameter of crimped filaments. The diameter of crimped filaments from 4.4.1.2.1 shall be determined as specified in H-S-951 for conformance with 3.2.1.2 or 3.2.1.3, as applicable.

4.4.1.4 Diameter of X-shaped filaments. Select 50 filaments at random from the filaments in 4.4.1.2.1. Measure the diameter of each filament as shown in figure 5 with a micrometer graduated and calibrated to read to 0.0001 inch. Care shall be taken not to distort or flatten the shape of the filament with micrometer pressure. Calculate the average value of the 50 filaments to the nearest 0.001 inch and check the result for conformance with 3.2.1.2 or 3.2.1.3, as applicable.

4.4.1.5 Wave amplitude and crimp frequency of filaments. The wave amplitude and crimp frequency of class 2 nylon and class 3 polypropylene filaments from 4.4.1.2.1 shall be determined by the applicable method of H-S-951 and the result checked for conformance with 3.2.1.2 or 3.2.1.3, as applicable.

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4.4.1.6 Melting point and specific gravity of filaments. The melting point and specific gravity of class 2 nylon and class 3 polypropylene filaments from 4.4.1.2.1 shall be determined by the applicable method in H-S-951 and the result checked for conformance with the applicable requirements in H-S-951.

4.4.1.7 Resistance to boiling water. Submerge the remaining fibers or filaments from 4.4.1.2.1 in a container of boiling water, 100° ± 1.1°C. (212° ± 2°F.). Allow the fibers or filaments to remain in the water, maintained at the boiling point, for not less than one hour, taking care that they are completely covered by the boiling water. At the end of the required time, examine the fibers or filaments for evidence of artificial color or in the case of black synthetic filaments for bleeding to determine conformance with 3.6.3. Dry the fibers or filaments in air for not less than one hour and examine for any change in stiffness to determine conformance with 3.6.3.

4.4.2 Denseness of fibers of filaments in brush head. Using a metal cylindrical rod 0.125 ± 0.001 inch in diameter, such as a drill rod, held so that it is parallel to the working surface formed by the fibers or filaments, attempt to insert the rod between tufts over the entire brush head, until it touches the wire or wood, as applicable. If the fibers or filaments are not sufficiently dense to prevent insertion of the rod as described above without touching the ends of the fibers or filaments or bending them aside, the brush shall be considered as having failed to meet the requirements in 3.4.1.1, 3.4.2.1 or 3.4.3.1, as applicable.

4.4.3 Tuft retention. For the purpose of testing twisted in-wire brushes, a specimen shall be a tuft, that is, all of the fibers or filaments held in one twist of two wire strands. For wire-bound tufts, the specimen shall consist of 100 fibers from one circular tuft. Five specimens from different locations on the brush head shall be tested. In the case of type I curved brush heads, at least two of the five tuft specimens shall be located in the leading end of the brush head. The test shall be as follows: Secure the handle in a suitable stationary clamping device and one end of the fibers or filaments in the specimen for test in another clamp so that they are evenly distributed and so as to prevent distortion and cutting. Care shall be taken to assure that both ends of the same fiber or filament are not held in the clamping device. By means of a spring scale or weight, apply a force of 8 ± 0.5 pounds in a direction perpendicular to the twisted wire strands to determine conformance with 3.7.

4.4.4 Resistance to chemicals.

4.4.4.1 Twenty-four hour immersion test. One duty A brush from each test unit shall be tested for resistance to sodium hypochlorite solution as specified in table V. One duty B brush from each test unit shall be tested for resistance to sodium hypochlorite solution and one from each test unit for resistance to sulfuric acid solution as specified in table V, by the following method: Suspend each brush for the required time in a container holding a sufficient quantity of the required solution to completely cover the brush head and at least 2 inches of the handle. The brush head shall not come into contact with the bottom or sides of the container. The temperature of the solution shall be 23 ± 2 C. (73.5 ± 3.6 F.) and shall be maintained at this temperature during the test. At the end of the required time (see table V), remove the brush from the solution, rinse it thoroughly in water, and examine for compliance with 3.8.1. The brush shall be allowed to dry for not less than 24 hours and then tested as described in 4.4.5, 4.4.6, and 4.4.7, as applicable.

4.4.5 Pull test. Hold the handle firmly in one hand and the brush part in the other and pull with any force capable of being exerted without mechanical aid to determine conformance with 3.8.2.

4.4.6 Turning test. Grasp the handle around the ferrule end with one hand and the brush head with the other hand. Twist the brush head rapidly clockwise and counterclockwise (approximately 10 degrees) ten times in each direction, taking care not to permanently distort the wire. Observe the evidence of loosening of wire within the handle and break-down of adhesive to determine conformance with 3.8.2.

4.4.7 Resistance of brush to impact. Drop the brush five times from a height of 10 feet onto a hard surface such as stone or concrete so that the handle strikes the surface each time. After each drop examine the brush for conformance with 3.8.2.

4.4.8 Density and hardness of plastic in type I brush handles. Prior to testing, two specimens from the plastic handle of one type I brush shall be conditioned for not less than 24 hours at a temperature of 23 ± 1.1 C. (73.5 ± 2 F.) and a relative humidity of 50 ± 4 percent, and tested under the same conditions. One specimen shall be tested in accordance with ASTM D 785 for Rockwell hardness and the other specimen in accordance with ASTM D 792 for density, to determine conformance with the applicable requirement in 3.2.3.

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4.4.9 Depth of wire in handle. With a suitable tool, cut the handle through its cross section 2-1/2 inches from the ferrule end for type I wood handles and 1-1/4 inches from the ferrule end for type I plastic brushes, as applicable, to determine compliance with the applicable requirement for depth of wire in the handle as specified in 3.4.1.2 or 3.4.1.3. One specimen from each test unit shall be tested.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Twelve brushes of like description shall be packaged in a snug-fitting fiberboard box conforming to style RSC, type CF or SF, class domestic of PPP-B-636. Brushes shall be arranged six wide, two deep, and reversed head to handle. Type I brushes shall be packaged on edge. A separator pad shall be placed between the layers of brushes. Closure shall be in accordance with the appendix to the box specification.

5.1.2 Level C. Brushes shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Seventy-two brushes of like description packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to V3c or V3s, style RSC of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.2 Level B. Unless otherwise specified, 72 brushes of like description, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF or SF, class domestic, of PPP-B-636, except that the bursting strength of the fiberboard shall be 200 pounds per square inch (p.s.i.). When specified, 12 brushes, packaged as specified in 5.1.1, shall require no overpacking. Closure shall be in accordance with the appendix of the container specification.

5.2.3 Level C. Brushes, packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe delivery at destination in compliance with National Motor Freight Classification or Uniform Freight Classification.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Types. Types I, II, and III brushes covered by this specification are intended primarily to clean wash basins, toilet bowls, and bedpans. Type IV brushes are intended primarily to clean cuspidors.

6.1.2 Duties. Duty A brushes are intended to be used in areas where the workload is light and exposure to cleaning agents is intermittent. Duty B brushes are intended to be used in areas where the work loads is heavy and they are continuously exposed to strong cleaning agents.

6.2 Ordering data. Purchasers should exercise any options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Duty, type, and class of brush required (see 1.2).
(c) When a preproduction sample is required (see 3.1).
(d) Applicable levels of packing and packaging.
(e) When plastic handles are required (see 3.4.1.2).

6.3 To prolong the life of sanitary brushes, it is recommended that they be rinsed in tap water after each exposure to cleaning agents and, when not in use, that they be suspended in air (not enclosed) so that the weight of the brush is not borne by the filaments, to promote drying, recovery of fibers and filaments, and to prevent rusting of wire.

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6.4 For information regarding ramin wood, see U. S. Forest Service Research Note FPL 0172, September 1967, "Seasoning and Handling of Ramin," by John M. McMillen, Technologist, Forest Products Laboratory, Forest Service, U. S. Department of Agriculture, Madison, Wisconsin 53705.

6.5 The following table gives a comparison of the classification in H-B-00481D and the preceeding Federal Specification H-B-481C.

Comparison of classification in H-B-00481D and H-B-481C.

Present classification H-B-00481D	Previous classification H-B-481C
Duty A. Light duty.	Not classified in this way.
Duty B. Heavy duty.	Not classified in this way.
Type I.	Type I.
Class 1. Tampico	Class 2. Tampico
Class 2. Nylon	Class 4. Nylon
Class 3. Polypropylene	Class 5. Polypropylene
Deleted.	Class 1. Hog bristle
Deleted.	Class 3. Horsetail hair
Type II.	Type II.
Class 1. Tampico	Class 2. Tampico
Type III.	Type III.
Class 2. Nylon	Class 4. Nylon
Class 3. Polypropylene	None.
Type IV. Ball brush head, wire bound	Type IV. Wire bound, ball head
Class 4. Palmetto	palmetto fiber

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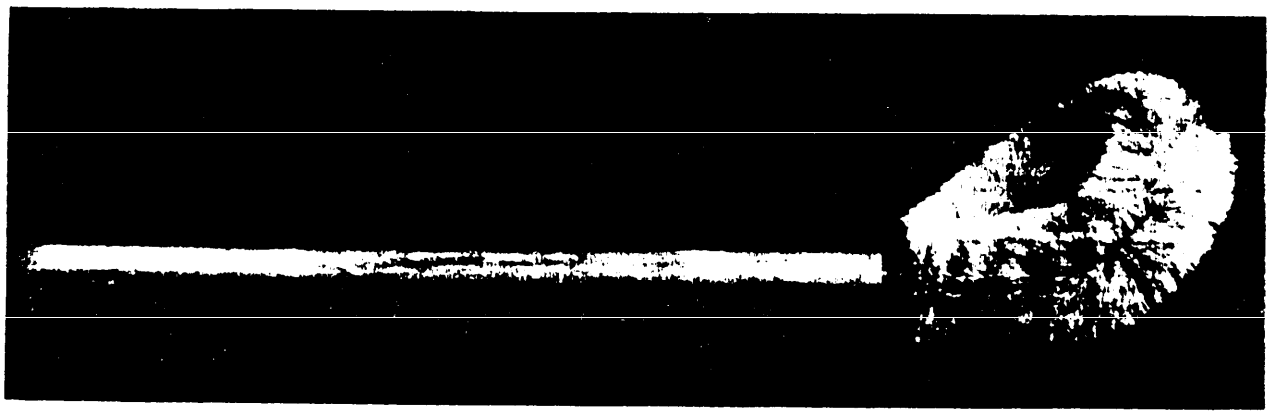


Figure 1

Type I- Twisted-in wire, curved brush head

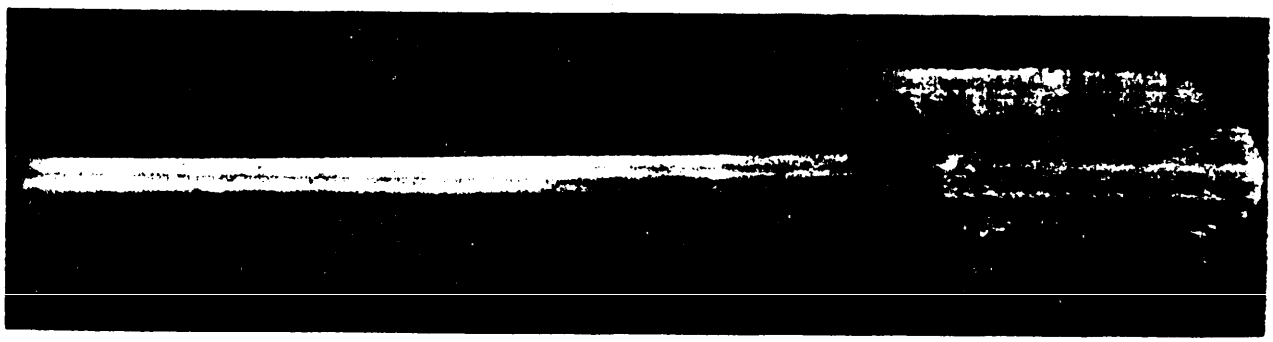


Figure 2

Type II- Spiral-wound, straight brush head

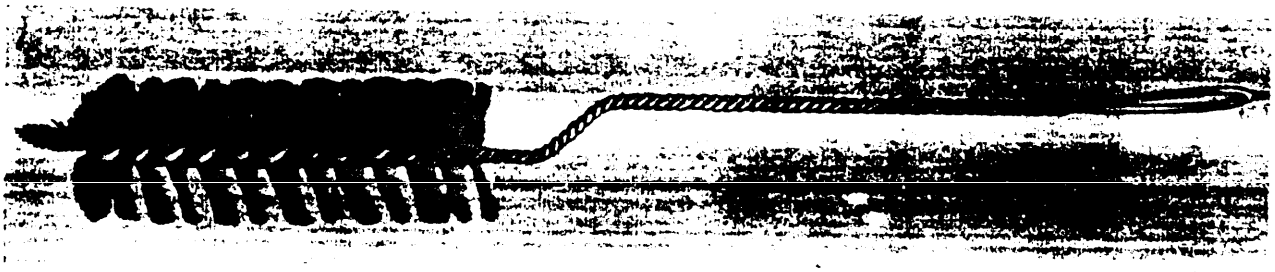


Figure 3

Type III- Twisted-in wire, straight brush head

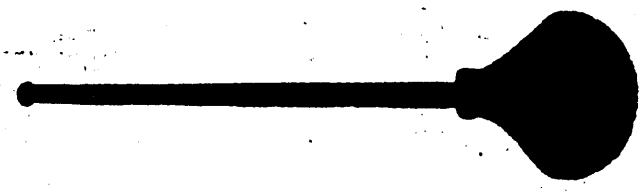
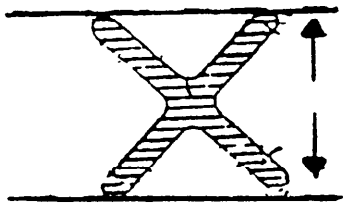


Figure 4

Type IV- Wire-bound, ball shaped brush head

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0.015 ± 0.001 (indicated diameter)

FIGURE 5
Magnified cross section of
X-shaped filaments

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, and mail to: General Services Administration, FSS (FMSO), Washington, D.C. 20406.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

H-B-00481D, BRUSHES, SANITARY

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

FORM GSA FEB 69 2200

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